The Kings School Mathematics Calculator Project Documentation

A guide for creating documentation for the Python program

Introduction

- Provide an overview of the project, its objectives, and its context.
- Explain the purpose and scope of the documentation.
- Outline the structure and organisation of the documentation.

Requirement Analysis

- Identify and describe the functional and non-functional requirements of the calculator program.
- Use tables, diagrams, or other tools to illustrate the requirements.
- Explain how the requirements were elicited and validated.
- Discuss any assumptions, constraints, or dependencies that affect the requirements.

Stage 1: Basic Calculator Functions

Flowchart for Stage 1

Insert the flowchart for the essential calculator functions here.

Pseudocode for Stage 1

Insert the pseudocode for the essential calculator functions here.

- Explain the challenges, issues, and solutions encountered in stage 1.

Stage 2: Advanced Calculator Functions

Flowchart for Stage 2

Insert the flowchart for the advanced calculator functions here.

Pseudocode for Stage 2

Insert the pseudocode for the advanced calculator functions here.

- Explain the challenges, issues, and solutions encountered in stage 2.

Stage 3: User Interface and Error Handling

Flowchart for Stage 3

Insert the flowchart for the user interface and error handling here.

Pseudocode for Stage 3

Insert the pseudocode for the user interface and error handling here.

- Explain the challenges, issues, and solutions encountered in stage 3.
- Provide the source code of the Python program with appropriate developer comments.
- Use consistent and clear coding style, indentation, and comments.
- Test the Python program for functionality, usability, and reliability.
- Provide the test cases, test data, expected results, and actual testing results.
- Explain the test results and any modifications made to the program.

Data Flow Diagram

Structure chart

Conclusion

- Summarise the main points and outcomes of the documentation.
- Evaluate the performance and quality of the calculator program.
- Reflect on the strengths and weaknesses of the documentation process.
- Provide recommendations and suggestions for future improvements.

References

- List the sources of information used in the documentation.
- Use a consistent and appropriate referencing style.